

Structural Emergence Research Program: Programmatic Scientific Claims and Research Directions

Toward Structurally Integrated Science, Meaning-Centered AI, and Emergence-Based Research Architectures

Author: Nikolai Ilin

Doctor of Sciences, Professor; Independent Researcher

E-mail: ilnik10@hotmail.com

License: CC BY 4.0

[ORCID: https://orcid.org/0009-0005-9286-7027](https://orcid.org/0009-0005-9286-7027)

Date: 2026

Abstract

This paper formulates the Structural Emergence Research Program as a theoretical and methodological contribution to the study of emergence, stability, meaning, artificial intelligence, and scientific integration. The paper does not present a closed doctrine, a final explanatory system, or a "theory of everything." Instead, it develops a programmatic research architecture: a structured set of provisional claims, conceptual distinctions, and research directions intended to support further formalization, empirical investigation, computational modeling, and interdisciplinary critique.

The motivation for the program is a structural tension within contemporary science. Scientific systems continue to expand in publication volume, technical specialization, data availability, and computational capacity. Yet informational expansion does not automatically produce conceptual integration. A growing body of work in the science of science has raised concerns about declining disruptiveness or reduced transformative novelty in some domains, while subsequent critiques have shown that such findings depend on measurement choices and disciplinary context. This debate is treated here not as a settled diagnosis, but as evidence of a broader problem: modern science is highly effective at producing local knowledge, but less systematically organized for synthesizing meaning across heterogeneous domains.

The paper proposes that emergence should be investigated as a threshold process in which interacting elements generate higher-order organizing structures. Stability is interpreted as dynamic persistence under transformation rather than as static equilibrium. Meaning is approached as structured integration rather than as an isolated property of symbols, signals, or statistical associations. Artificial intelligence is therefore discussed not merely as a tool for prediction or text generation, but as a possible infrastructure for meaning-centered scientific integration, provided that future systems develop stronger mechanisms of contextual stability, semantic organization, and cross-domain structural mapping.

The paper systematizes twenty-eight programmatic claims across four domains: science and knowledge systems; emergence and structural dynamics; artificial intelligence and meaning; and stability, solitons, and cosmological interpretation. Each claim is framed as a research thesis rather than as an established conclusion. The resulting contribution is a self-contained agenda for structurally integrated science: a mode of inquiry focused on relations, constraints, thresholds, persistence, and meaning-producing organization across complex systems.

Keywords: structural emergence; emergence theory; philosophy of science; scientific methodology; interdisciplinary research; meaning-centered artificial intelligence; cognitive

architectures; gestalt cognition; semantic integration; nonlinear dynamics; soliton theory; structural stability; complexity science.

1. Introduction

Modern science has achieved an unprecedented capacity for producing information. Scientific publication volume has grown dramatically, computational resources have expanded, and increasingly specialized research communities now investigate phenomena at levels of precision that would have been impossible in earlier periods. These developments have transformed the scope and power of scientific inquiry.

At the same time, informational growth creates a structural problem. More publications, datasets, models, and technical procedures do not automatically generate deeper conceptual integration. A scientific system may become extremely productive in local terms while remaining poorly optimized for cross-domain synthesis. It may generate large quantities of specialized knowledge without developing the architectures needed to connect that knowledge into coherent explanatory structures.

This concern appears in recent debates about the organization and productivity of science. Park, Leahey, and Funk (2023) argue that papers and patents have become less disruptive over time, using citation-based measures to estimate whether later work consolidates or destabilizes prior knowledge. Their conclusions have generated substantial discussion, and later analyses, such as Macher, Rutzer, and Weder (2024), indicate that measurement bias and field-specific factors may affect at least some interpretations of declining disruptiveness, especially in patent analysis. The present paper does not depend on accepting any single bibliometric conclusion. Rather, it treats the debate as an indication of a deeper methodological issue: scientific productivity cannot be evaluated only by output volume. It also depends on whether research systems generate new organizing principles, integrate fragmented results, and transform the structure of explanation.

The difficulty is especially visible in problems that cross levels of organization. Emergence, cognition, artificial intelligence, social complexity, biological organization, climate systems, and cosmological interpretation cannot be adequately understood through isolated disciplinary fragments alone. They require analysis of relations, constraints, thresholds, feedback, stability, and meaning formation. These are structural problems, not merely domain-specific problems.

The Structural Emergence Research Program is proposed as a response to this situation. It does not reject reductionist analysis, mathematical formalization, empirical testing, or disciplinary specialization. On the contrary, these remain indispensable. The program argues, however, that scientific explanation also requires explicit attention to structural organization: the relations among components, the constraints that shape possible interactions, the thresholds at which new levels of order appear, and the mechanisms by which meaning and stability are maintained across change.

The guiding thesis is that complex systems cannot be understood only by decomposing them into isolated elements. They must also be studied as organized systems in which relations, patterns, and constraints generate higher-order forms. Emergence, in this view, is not an appeal to mystery. It is a research problem: under what conditions do interactions produce new organizational levels, and how can those levels be modeled, measured, tested, and compared?

The paper has four objectives. First, it defines the methodological status of the Structural Emergence Research Program as a theoretical and methodological article rather than as a completed theory. Second, it describes the structural architecture of the program. Third, it

systematizes twenty-eight programmatic claims derived from the program. Fourth, it identifies research directions in meaning-centered AI, gestalt-level cognitive architectures, predictive emergence modeling, structural mapping systems, and integrative scientific platforms.

The intended contribution is programmatic but not merely speculative. The claims are presented as research theses that require further elaboration. They are designed to become objects of formal modeling, computational simulation, empirical investigation, and critical evaluation. The value of the paper lies not in claiming finality, but in making a coherent research space explicit.

2. The Structural Emergence Research Program

2.1. Origin and Scope

The Structural Emergence Research Program developed from a set of recurring questions across emergence theory, nonlinear dynamics, cognitive science, artificial intelligence, cosmology, and scientific methodology. What distinguishes emergence from simple aggregation? How do stable forms persist in nonlinear systems? How does meaning arise from relations rather than from isolated symbols? Why do some systems develop coherent higher-order organization while others remain fragmented or unstable? How can science organize meaning across increasingly specialized domains?

Although these questions appear to belong to separate fields, they share a common structural form. Each concerns a transition from multiplicity to organization: from elements to systems, from local interactions to global patterns, from signals to meaning, from instability to persistence, and from disciplinary fragments to integrative explanation.

The program therefore treats emergence not as a single phenomenon restricted to one field, but as a family of structural transitions. It also treats stability and meaning as related problems. A structure becomes scientifically significant when it not only appears, but persists; and meaning becomes scientifically significant when it is not merely asserted, but organized and stabilized within a system of relations.

2.2. From Research Topics to Research Architecture

A research architecture is more than a list of topics. It identifies the relations among problems, concepts, hypotheses, and methods. The Structural Emergence Research Program organizes its materials around recurring structural motifs: thresholds, constraints, third-element formation, resonance, persistence, semantic integration, and cross-domain mapping.

The program does not claim that all domains are identical. Physical solitons, biological self-organization, cognitive gestalt perception, AI semantic integration, and scientific methodology are not the same phenomenon. However, they may display comparable organizational problems. The task is to identify when such comparisons are productive, when they are merely metaphorical, and when they can be formalized into testable models.

This distinction is important. Interdisciplinary research often fails when analogy is treated as proof. The present program treats cross-domain correspondences as hypotheses. They are valuable only if they generate clearer definitions, better models, new predictions, or improved explanatory integration.

2.3. Three Core Cycles

The program can be organized into three core cycles.

Cycle A concerns emergence. It investigates how interactions cross thresholds and produce higher-order organizing structures. This cycle focuses on the difference between aggregation and emergence, the role of constraints, and the possibility of modeling emergence through the appearance of a structurally organizing third element.

Cycle B concerns stability. It investigates how emergent structures persist under transformation. Soliton-like dynamics are used as a reference model because solitons demonstrate localized stability in nonlinear media. The claim is not that all persistent structures are literal solitons. The claim is that soliton-like persistence provides a useful conceptual and mathematical analogy for studying dynamic coherence.

Cycle C concerns meaning and artificial intelligence. It investigates meaning as structured integration across context, memory, function, and interpretation. This cycle argues that AI systems require more than statistical fluency if they are to support robust meaning-centered reasoning. They require architectures that maintain semantic stability across context shifts and organize information into coherent wholes.

2.4. Meta-Level Integration

The three cycles require a meta-level integrative layer. If emergence, stability, and meaning are studied only in isolation, the research program would reproduce the fragmentation it seeks to address. The integrative layer therefore concerns science itself: how knowledge is organized, how cross-domain correspondences are identified, and how AI-supported systems might help science move from informational accumulation toward structural synthesis.

Structurally integrated science does not mean replacing existing disciplines. It means designing methods, representations, and institutions that allow relations among disciplines to become visible and scientifically usable.

3. Methodological Status of the Claims

The claims formulated in this paper should not be interpreted as universally established conclusions, completed proofs, or final theoretical laws. They are programmatic scientific theses. Their primary function is to organize inquiry by identifying relations, distinctions, and research directions that may be scientifically productive.

A programmatic claim occupies a middle position between speculation and established result. It is not yet a proven conclusion, but it is not an arbitrary assertion. It is a disciplined research proposition that can guide formalization, testing, and critique. Such claims are common in emerging scientific programs, where conceptual organization often precedes full mathematical or empirical closure.

Several claims in this paper require substantial development. The idea that emergence can be modeled through a structurally organizing third element requires clearer definitions of element, relation, threshold, organization, constraint, and stability. The proposal that meaning-centered AI requires gestalt-level semantic integration requires operational metrics for semantic coherence, contextual robustness, and interpretive continuity. The use of soliton-like models outside physics requires careful distinction between literal mathematical solitons, partial formal analogues, and broader structural metaphors.

The program therefore adopts four methodological commitments.

First, it treats emergence as a research problem rather than as a residual category. Emergence should not mean "what cannot currently be reduced." It should refer to identifiable structural conditions under which new organizational levels appear.

Second, it treats cross-domain analogy as heuristic rather than evidential by itself. Analogies are useful when they generate testable hypotheses or clearer structural distinctions. They are not sufficient as proof.

Third, it distinguishes strong claims from research directions. Some claims may eventually become formal models; others may remain conceptual tools; others may be rejected. The program is designed to allow revision.

Fourth, it avoids both reductive closure and speculative holism. Reductionism risks ignoring relational organization; speculative holism risks invoking emergence or meaning without operational constraints. The research program seeks a middle path in which higher-order structures are investigated rigorously without being reduced prematurely to isolated components.

It is not a manifesto, not a closed doctrine, and not a final theory of all complex systems.

4. Structural Architecture of the Program

The architecture of the program can be summarized through four interacting layers: emergence, stability, meaning, and integration.

The emergence layer studies the formation of higher-order organization. Its central concern is the transition from interaction to structure. A system becomes emergent when its elements are organized by relations that generate new properties, constraints, or functions not adequately captured by the elements in isolation.

The stability layer studies persistence. Emergent structures matter scientifically when they can maintain coherence across time, perturbation, and transformation. Stability is therefore not treated merely as static equilibrium. It is understood as dynamic persistence.

The meaning layer studies structured interpretation. Meaning arises when elements become organized into context-sensitive relations that determine relevance, function, and significance. This applies to cognition, language, AI, and scientific explanation.

The integration layer studies the organization of knowledge itself. It asks how scientific fields can identify structural correspondences without collapsing disciplinary differences. It also asks how AI systems may assist the formation of integrative scientific architectures.

These layers are mutually reinforcing. Emergence requires stability if new structures are to persist. Stability requires organization if persistence is to be more than inertia. Meaning requires integration if isolated signals are to become interpretable. Science requires all three if it is to move from information accumulation toward explanatory synthesis.

5. Programmatic Scientific Claims

The twenty-eight claims below are organized into four thematic domains. Each claim is deliberately formulated as a research thesis rather than as a final conclusion.

Part I. Science and Knowledge Systems

Claim 1. Structural fragmentation is becoming a primary limitation of scientific progress.

Modern science increasingly optimizes publication activity, technical specialization, and local problem solving, while less consistently optimizing conceptual integration across domains. The central limitation may no longer be lack of information, but difficulty organizing information into coherent explanatory structures.

Claim 2. Scientific productivity should be evaluated by integrative capacity as well as output volume.

Publication counts, citation metrics, and technical outputs cannot fully measure whether a scientific system produces transformative explanatory architectures. A mature evaluation of science should include the capacity for synthesis, conceptual reorganization, and cross-domain integration.

Claim 3. Excessive specialization can produce epistemic isolation.

Specialization is indispensable for technical precision. However, when specialization is not balanced by integrative mechanisms, it can produce isolated models that do not communicate effectively across fields.

Claim 4. Scientific knowledge increasingly resembles a distributed archive rather than an integrated cognitive system.

The accumulation of disconnected scientific outputs can produce informational abundance without proportional growth in organized meaning. Scientific modernization therefore requires architectures of synthesis, not only repositories of information.

Claim 5. Interdisciplinary science requires structural translation.

Interdisciplinarity is not achieved merely by combining topics from multiple fields. It requires identifying relations, constraints, and organizational patterns that can be translated across domains while preserving domain-specific rigor.

Claim 6. Knowledge architecture is becoming a central scientific problem.

As scientific information grows beyond human-scale synthesis, the design of systems for organizing, linking, evaluating, and integrating knowledge becomes a core methodological task.

Claim 7. AI can become an instrument of scientific integration.

Artificial intelligence may assist science not only by accelerating computation or literature search, but by identifying structural correspondences across disciplines. This requires AI systems designed for meaning integration rather than merely output generation.

Claim 8. Research programs should be evaluated as architectures, not only as isolated publications.

A set of papers may have scientific value when treated as an interconnected research architecture. Program-level evaluation can reveal relations among claims, methods, and concepts that are not visible when works are assessed only individually.

Part II. Emergence and Structural Dynamics

Claim 9. Emergence is a threshold phenomenon.

Emergence occurs when interacting elements cross a threshold at which a new organizing level appears. The threshold may involve feedback, constraint formation, nonlinearity, phase transition, or semantic closure, depending on the domain.

Claim 10. Emergence requires more than aggregation.

A system is not emergent merely because it contains many parts. Emergence requires relational organization that changes the behavior, function, or identity of the system as a whole.

Claim 11. The structurally organizing third element is a candidate mechanism of emergence.

When interacting elements generate a new stabilizing relation, constraint, context, or functional structure, a third element appears that reorganizes the original interaction. This concept provides a candidate model for the transition from interaction to organization.

Claim 12. Higher-order structures can exert downward constraint without violating lower-level causality.

Emergent structures may influence lower-level processes by constraining possible interactions, stabilizing patterns, and altering feedback conditions. This is not a rejection of lower-level causality but an account of multi-level organization.

Claim 13. Structural emergence requires both relation and persistence.

A transient pattern may be interesting, but an emergent structure becomes scientifically significant when it persists long enough to shape system behavior. Emergence and stability are therefore analytically connected.

Claim 14. Predictive emergence models should identify conditions under which new organization becomes likely.

Emergence research should move beyond retrospective description. It should seek indicators that predict when local interactions may generate new organizational levels.

Part III. Artificial Intelligence and Meaning

Claim 15. Meaning is a structural phenomenon.

Meaning does not reside in isolated symbols, data points, or tokens. It arises from organized relations among context, function, memory, perception, action, and interpretation.

Claim 16. Statistical fluency is not equivalent to semantic integration.

AI systems may produce fluent language while lacking stable meaning structures. The difference between plausible output and robust understanding should be treated as a central research problem.

Claim 17. Gestalt-level cognition is necessary for robust meaning.

Meaning depends on the organization of parts into coherent wholes. A system that processes isolated units without whole-form integration will remain limited in its capacity for contextual interpretation.

Claim 18. Context is constitutive of meaning.

A sign, symbol, or signal becomes meaningful within a structured field of relations. Context is not merely additional information; it is part of the formation of meaning.

Claim 19. AI systems require semantic stability mechanisms.

For AI systems to maintain reliable meaning across changing contexts, they require mechanisms that preserve semantic coherence under variation, ambiguity, and perturbation.

Claim 20. Memory must be structurally integrated rather than merely stored.

Intelligence requires not only access to stored information, but organization of memory into relational structures that support interpretation, continuity, and learning.

Claim 21. AI alignment requires meaning-level interpretation.

Alignment cannot be reduced to output filtering or rule-following alone. It requires systems capable of interpreting goals, contexts, consequences, and human meanings at structural levels.

Claim 22. Meaning-centered AI requires architectural innovation, not only scaling.

Increasing model size may improve performance, but scaling alone does not guarantee semantic grounding, contextual stability, or integrative understanding. Future AI research should therefore investigate architecture-centered approaches.

Claim 23. AI systems may help detect emergent structures if designed for structural mapping.

AI can potentially identify cross-domain patterns of emergence, stability, and semantic organization if equipped with appropriate representations and evaluation criteria.

Part IV. Stability, Solitons, and Cosmological Interpretation

Claim 24. Stability is an active dynamic process.

Many stable systems persist not by remaining unchanged, but by maintaining coherent organization through adaptation, feedback, and transformation.

Claim 25. Soliton-like models provide a structural language for persistence.

Solitons offer a model of localized dynamic stability in nonlinear systems. Soliton-like concepts may help formulate broader models of persistence, provided that literal physical claims are distinguished from analogical or formal extensions.

Claim 26. Resonance may function as a mechanism of stabilization.

Systems may become stable when internal dynamics and external conditions enter into coherence, feedback, or resonance-like relation. The exact meaning of resonance must be formalized separately in each domain.

Claim 27. Cosmological structures may benefit from emergence and stability frameworks.

Some problems in cosmology may be clarified through attention to phase transitions, nonlinear stability, field structure, and emergent organization. This claim is exploratory and requires domain-specific mathematical development.

Claim 28. Structurally integrated science may represent a necessary next stage of scientific modernization.

The next stage of scientific development may depend less on further accumulation of isolated findings and more on integrative cognitive architectures capable of organizing meaning across heterogeneous domains.

6. Cross-Claim Integration

The twenty-eight claims are not independent slogans. They are connected by recurring structural motifs.

6.1. Threshold Dynamics

Threshold dynamics links emergence, cognition, AI, and science studies. A system changes qualitatively when interactions cross a boundary and produce a new organizational level. In science, this may appear as paradigm formation. In cognition, it may appear as gestalt perception. In AI, it may appear as a transition from pattern association to semantic organization.

6.2. Constraint and Organization

Constraints are central to emergence. They do not merely limit behavior; they make new forms of organization possible. A system without constraints may remain chaotic or indeterminate. A system with appropriate constraints can form stable patterns, functional roles, and meaningful relations.

6.3. Persistence Through Transformation

Persistence is not the absence of change. Many structures persist by transforming in organized ways. Biological organisms, cognitive identities, scientific programs, and semantic structures all maintain continuity through controlled change. This motif links stability theory with meaning and scientific methodology.

6.4. Meaning as Structured Relation

Meaning appears wherever elements acquire roles within an organized field of relations. This applies to language, cognition, AI, and scientific explanation. A theory is meaningful not because it contains many propositions, but because it organizes them into an explanatory structure.

6.5. Integration as a Scientific Function

Integration is not a secondary activity performed after "real" research. It is itself a scientific function. The ability to organize knowledge across domains may become increasingly central as scientific information expands.

7. Research Directions

7.1. Meaning-Centered AI

Meaning-centered AI should investigate architectures that integrate context, memory, perception, action, and semantic constraint. Key research questions include: How can semantic stability be measured? What distinguishes fluent output from meaning-level understanding? Can AI systems maintain coherent interpretations across context shifts? What forms of grounding are necessary for robust interpretation? How can systems detect contradictions structurally rather than only syntactically?

Possible methods include structured memory systems, semantic graph models, hybrid cognitive architectures, embodied or environment-linked learning systems, and evaluation benchmarks focused on meaning persistence.

7.2. Gestalt-Level Cognitive Architectures

Gestalt-level architectures should study cognition as whole-form organization. Such architectures would not treat perception, memory, and action as isolated modules, but as mutually constraining components of an integrated system. Research tasks include modeling transitions from local features to global wholes, representing context as constitutive of meaning, and comparing human gestalt cognition with machine representation.

7.3. Structural Mapping Systems

Structural mapping systems would identify recurring organizational patterns across domains. They may combine AI-assisted literature analysis, graph modeling, formal ontologies, and human theoretical interpretation. Outputs could include cross-domain matrices, maps of emergence mechanisms, databases of stability patterns, and semantic integration graphs.

7.4. Predictive Emergence Models

Predictive emergence models would move emergence research from retrospective description toward prospective analysis. Instead of identifying emergence only after a new pattern appears, such models would ask whether threshold conditions can be specified in advance. Research questions include whether feedback, constraint, resonance, and network measures can estimate emergent potential.

7.5. Integrative Scientific Platforms

Future scientific platforms should not merely store publications. They should organize knowledge. Such platforms may include AI-assisted synthesis tools, claim-to-evidence matrices, cross-disciplinary translation layers, structural concept maps, and tools for detecting conceptual fragmentation and integration opportunities.

8. Implications for the Future of Science

The strongest implication of the program is that the future of science may depend less on further expansion of isolated disciplines and more on the development of integrative cognitive architectures capable of organizing meaning across domains.

This does not imply that disciplines should disappear. Disciplines provide rigor, expertise, and technical depth. However, many contemporary problems are structurally distributed across levels of organization. Climate systems, artificial intelligence, cognition, biological complexity, social instability, and cosmological interpretation cannot be addressed adequately by isolated fragments alone.

The proposed modernization of science has three dimensions. First, science requires conceptual integration. Second, science requires structural methodology: tools for identifying relations, constraints, thresholds, stability, and emergent organization. Third, science increasingly requires AI-supported knowledge architectures that can assist with synthesis at scales beyond unaided human cognition.

AI will not automatically solve this problem. If AI systems remain optimized primarily for surface-level output generation, they may intensify fragmentation by producing more text without improving understanding. Meaning-centered AI is therefore both an application of the program and a condition for its future development.

The next stage of scientific development may involve a double emergence: the emergence of more integrated scientific knowledge architectures and the emergence of AI systems capable of participating responsibly in meaning-centered integration.

9. Limitations

This paper has several limitations. First, many claims remain programmatic and require formalization. The concepts of threshold, third element, semantic stability, and structural persistence must be specified more rigorously in future work.

Second, cross-domain integration carries the risk of overgeneralization. Structural similarities across physics, biology, cognition, AI, and science studies may be useful, but they must not be treated as identity claims. Each domain requires its own evidence, methods, and constraints.

Third, the paper does not provide empirical validation of the full research program. Its contribution is theoretical and methodological. It organizes a research agenda, but the agenda must be tested through future mathematical, computational, and empirical work.

Fourth, the use of soliton-like models outside physics should be interpreted cautiously. Some extensions may become formal analogues; others may remain heuristic metaphors. The distinction must be made explicit in subsequent research.

Finally, the program depends on future implementation. Its value will be determined by whether it generates better models, clearer hypotheses, useful AI systems, productive interdisciplinary research, or improved scientific integration.

10. Conclusion

This paper has formulated the Structural Emergence Research Program as a theoretical and methodological research architecture. Its central claim is that emergence, stability, meaning, and scientific integration are not isolated problems. They are different expressions of a deeper structural question: how do distributed elements become organized into coherent higher-order forms?

The paper has argued that contemporary science faces a structural challenge. It is increasingly capable of generating information, but less systematically optimized for organizing meaning across fragmented domains. In response, the program proposes structurally integrated science as a methodological orientation focused on relations, constraints, thresholds, persistence, and meaning-producing organization.

The twenty-eight claims presented here are not final conclusions. They are research theses intended to guide future formalization, testing, revision, and critique. Some may become mathematical models; others may function as philosophical distinctions; others may inform AI architecture, scientific infrastructure, or interdisciplinary methodology.

The most important contribution of the paper is not that it solves the problems it identifies. Rather, it makes those problems visible as part of one research space. If successful, the Structural Emergence Research Program may contribute to a future science that is not only more productive, but more integrated; not only more informationally powerful, but more capable of organizing meaning.

Data and Ethics Statements

Data availability: This is a theoretical and methodological article. No empirical dataset was generated or analyzed for this manuscript.

Ethics statement: This article does not report research involving human participants, animals, clinical intervention, or confidential personal data.

References

- Anderson, P. W. (1972). More is different: Broken symmetry and the nature of the hierarchical structure of science. *Science*, 177(4047), 393-396. <https://doi.org/10.1126/science.177.4047.393>
- Ashby, W. R. (1956). *An Introduction to Cybernetics*. Chapman & Hall.
- Barsalou, L. W. (2008). Grounded cognition. *Annual Review of Psychology*, 59, 617-645. <https://doi.org/10.1146/annurev.psych.59.103006.093639>
- Bedau, M. A. (1997). Weak emergence. *Philosophical Perspectives*, 11, 375-399.
- Dauxois, T., & Peyrard, M. (2006). *Physics of Solitons*. Cambridge University Press.

- Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127-138. <https://doi.org/10.1038/nrn2787>
- Harnad, S. (1990). The symbol grounding problem. *Physica D: Nonlinear Phenomena*, 42(1-3), 335-346. [https://doi.org/10.1016/0167-2789\(90\)90087-6](https://doi.org/10.1016/0167-2789(90)90087-6)
- Holland, J. H. (1998). *Emergence: From Chaos to Order*. Oxford University Press.
- Kauffman, S. A. (1993). *The Origins of Order: Self-Organization and Selection in Evolution*. Oxford University Press.
- Laughlin, R. B., & Pines, D. (2000). The theory of everything. *Proceedings of the National Academy of Sciences*, 97(1), 28-31. <https://doi.org/10.1073/pnas.97.1.28>
- Macher, J. T., Rutzer, C., & Weder, R. (2024). Is there a secular decline in disruptive patents? Correcting for measurement bias. *Research Policy*, 53(5), 104992. <https://doi.org/10.1016/j.respol.2024.104992>
- Mitchell, M. (2009). *Complexity: A Guided Tour*. Oxford University Press.
- Newell, A. (1990). *Unified Theories of Cognition*. Harvard University Press.
- Park, M., Leahey, E., & Funk, R. J. (2023). Papers and patents are becoming less disruptive over time. *Nature*, 613, 138-144. <https://doi.org/10.1038/s41586-022-05543-x>
- Simon, H. A. (1962). The architecture of complexity. *Proceedings of the American Philosophical Society*, 106(6), 467-482.
- Varela, F. J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press.